# MSG No. 208.47 $P_I4_232$ [ Type IV, cubic ]

Table 1: Wyckoff site: 2a, site symmetry: 4'32'

| No. | position | mapping |
|---|---|---|
| 1 | [0, 0, 0] | [1,8,9,10,17,18,19,20,21,22,23,24,26,27,28,29,30,31,35,36,37,38,39,40] |
| 2 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | [2,3,4,5,6,7,11,12,13,14,15,16,25,32,33,34,41,42,43,44,45,46,47,48] |

Table 2: Wyckoff site: 6b, site symmetry: 4'2.2'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | [1,8,9,10,26,27,37,38] |
| 2 | $\left[\frac{1}{2}, 0, 0\right]$ | [2,3,13,14,25,32,33,34] |
| 3 | $\left[0, 0, \frac{1}{2}\right]$ | [4,5,15,16,42,43,45,47] |
| 4 | $\left[0, \frac{1}{2}, 0\right]$ | [6,7,11,12,41,44,46,48] |
| 5 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | [17,20,22,24,30,31,35,36] |
| 6 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | [18,19,21,23,28,29,39,40] |

Table 3: Wyckoff site: 8c, site symmetry: .32

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | [1,12,14,16,17,18] |
| 2 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | [2,7,9,15,19,24] |
| 3 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | [3,4,10,11,20,21] |
| 4 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | [5,6,8,13,22,23] |
| 5 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | [25,36,38,40,41,42] |
| 6 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | [26,31,33,39,43,48] |
| 7 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | [27,28,34,35,44,45] |
| 8 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | [29,30,32,37,46,47] |

Table 4: Wyckoff site: 12d, site symmetry: 2.22

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{2}, 0\right]$ | [1,8,13,14] |
| 2 | $\left[\frac{3}{4}, \frac{1}{2}, 0\right]$ | [2,3,9,10] |
| 3 | $\left[\frac{1}{2}, 0, \frac{1}{4}\right]$ | [4,16,18,21] |
| 4 | $\left[\frac{1}{2}, 0, \frac{3}{4}\right]$ | [5,15,19,23] |
| 5 | $\left[0, \frac{3}{4}, \frac{1}{2}\right]$ | [6,11,20,22] |
| 6 | $\left[0, \frac{1}{4}, \frac{1}{2}\right]$ | [7,12,17,24] |
| 7 | $\left[\frac{3}{4}, 0, \frac{1}{2}\right]$ | [25,32,37,38] |
| 8 | $\left[\frac{1}{4}, 0, \frac{1}{2}\right]$ | [26,27,33,34] |
| 9 | $\left[0, \frac{1}{2}, \frac{3}{4}\right]$ | [28,40,42,45] |
| 10 | $\left[0, \frac{1}{2}, \frac{1}{4}\right]$ | [29,39,43,47] |
| 11 | $\left[\frac{1}{2}, \frac{1}{4}, 0\right]$ | [30,35,44,46] |
| 12 | $\left[\frac{1}{2}, \frac{3}{4}, 0\right]$ | [31,36,41,48] |

Table 5: Wyckoff site: `12e`, site symmetry: `4'..`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ 0]$ | `[1,8,26,27]` |
| 2 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[2,3,25,32]` |
| 3 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-x\right]$ | `[4,16,43,47]` |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2},\ x+\frac{1}{2}\right]$ | `[5,15,42,45]` |
| 5 | $\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | `[6,11,41,48]` |
| 6 | $\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | `[7,12,44,46]` |
| 7 | $[-x,\ 0,\ 0]$ | `[9,10,37,38]` |
| 8 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[13,14,33,34]` |
| 9 | $[0,\ x,\ 0]$ | `[17,24,30,35]` |
| 10 | $[0,\ 0,\ x]$ | `[18,21,29,39]` |
| 11 | $[0,\ 0,\ -x]$ | `[19,23,28,40]` |
| 12 | $[0,\ -x,\ 0]$ | `[20,22,31,36]` |

Table 6: Wyckoff site: `16f`, site symmetry: `.3.`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ x,\ x]$ | `[1,17,18]` |
| 2 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$ | `[2,7,15]` |
| 3 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$ | `[3,4,11]` |
| 4 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$ | `[5,6,13]` |
| 5 | $[x,\ -x,\ -x]$ | `[8,22,23]` |
| 6 | $[-x,\ x,\ -x]$ | `[9,19,24]` |
| 7 | $[-x,\ -x,\ x]$ | `[10,20,21]` |
| 8 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$ | `[12,14,16]` |
| 9 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$ | `[25,41,42]` |
| 10 | $[x,\ -x,\ x]$ | `[26,31,39]` |
| 11 | $[x,\ x,\ -x]$ | `[27,28,35]` |
| 12 | $[-x,\ x,\ x]$ | `[29,30,37]` |
| 13 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$ | `[32,46,47]` |
| 14 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$ | `[33,43,48]` |
| 15 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$ | `[34,44,45]` |
| 16 | $[-x,\ -x,\ -x]$ | `[36,38,40]` |

Table 7: Wyckoff site: `24g`, site symmetry: `2..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{1}{2},\ 0\right]$ | `[1,8]` |
| 2 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$ | `[2,3]` |
| 3 | $\left[\frac{1}{2},\ 0,\ \frac{1}{2}-x\right]$ | `[4,16]` |
| 4 | $\left[\frac{1}{2},\ 0,\ x+\frac{1}{2}\right]$ | `[5,15]` |
| 5 | $\left[0,\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | `[6,11]` |
| 6 | $\left[0,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | `[7,12]` |
| 7 | $\left[-x,\ \frac{1}{2},\ 0\right]$ | `[9,10]` |
| 8 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$ | `[13,14]` |

*continued ...*

Table 7

| No. | position | mapping |
|---|---|---|
| 9 | $\left[0,\, x,\, \frac{1}{2}\right]$ | [17,24] |
| 10 | $\left[\frac{1}{2},\, 0,\, x\right]$ | [18,21] |
| 11 | $\left[\frac{1}{2},\, 0,\, -x\right]$ | [19,23] |
| 12 | $\left[0,\, -x,\, \frac{1}{2}\right]$ | [20,22] |
| 13 | $\left[x+\frac{1}{2},\, 0,\, \frac{1}{2}\right]$ | [25,32] |
| 14 | $\left[x,\, 0,\, \frac{1}{2}\right]$ | [26,27] |
| 15 | $\left[0,\, \frac{1}{2},\, -x\right]$ | [28,40] |
| 16 | $\left[0,\, \frac{1}{2},\, x\right]$ | [29,39] |
| 17 | $\left[\frac{1}{2},\, x,\, 0\right]$ | [30,35] |
| 18 | $\left[\frac{1}{2},\, -x,\, 0\right]$ | [31,36] |
| 19 | $\left[\frac{1}{2}-x,\, 0,\, \frac{1}{2}\right]$ | [33,34] |
| 20 | $\left[-x,\, 0,\, \frac{1}{2}\right]$ | [37,38] |
| 21 | $\left[\frac{1}{2},\, x+\frac{1}{2},\, 0\right]$ | [41,48] |
| 22 | $\left[0,\, \frac{1}{2},\, x+\frac{1}{2}\right]$ | [42,45] |
| 23 | $\left[0,\, \frac{1}{2},\, \frac{1}{2}-x\right]$ | [43,47] |
| 24 | $\left[\frac{1}{2},\, \frac{1}{2}-x,\, 0\right]$ | [44,46] |

Table 8: Wyckoff site: 24h, site symmetry: ..2'

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\, y,\, y]$ | [1,37] |
| 2 | $\left[\frac{1}{2},\, \frac{1}{2}-y,\, y+\frac{1}{2}\right]$ | [2,34] |
| 3 | $\left[\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-y\right]$ | [3,33] |
| 4 | $\left[y+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}\right]$ | [4,42] |
| 5 | $\left[\frac{1}{2}-y,\, y+\frac{1}{2},\, \frac{1}{2}\right]$ | [5,43] |
| 6 | $\left[\frac{1}{2}-y,\, \frac{1}{2},\, y+\frac{1}{2}\right]$ | [6,44] |
| 7 | $\left[y+\frac{1}{2},\, \frac{1}{2},\, y+\frac{1}{2}\right]$ | [7,41] |
| 8 | $[0,\, -y,\, -y]$ | [8,38] |
| 9 | $[0,\, y,\, -y]$ | [9,27] |
| 10 | $[0,\, -y,\, y]$ | [10,26] |
| 11 | $\left[y+\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}-y\right]$ | [11,46] |
| 12 | $\left[\frac{1}{2}-y,\, \frac{1}{2},\, \frac{1}{2}-y\right]$ | [12,48] |
| 13 | $\left[\frac{1}{2},\, y+\frac{1}{2},\, y+\frac{1}{2}\right]$ | [13,25] |
| 14 | $\left[\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-y\right]$ | [14,32] |
| 15 | $\left[y+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$ | [15,47] |
| 16 | $\left[\frac{1}{2}-y,\, \frac{1}{2}-y,\, \frac{1}{2}\right]$ | [16,45] |
| 17 | $[y,\, 0,\, y]$ | [17,31] |
| 18 | $[y,\, y,\, 0]$ | [18,28] |
| 19 | $[-y,\, y,\, 0]$ | [19,29] |
| 20 | $[-y,\, 0,\, y]$ | [20,30] |
| 21 | $[-y,\, -y,\, 0]$ | [21,40] |
| 22 | $[y,\, 0,\, -y]$ | [22,35] |
| 23 | $[y,\, -y,\, 0]$ | [23,39] |
| 24 | $[-y,\, 0,\, -y]$ | [24,36] |

Table 9: Wyckoff site: `24i`, site symmetry: `..2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\ y,\ \frac{1}{2}-y\right]$ | `[1,14]` |
| 2 | $\left[\frac{3}{4},\ y,\ y+\frac{1}{2}\right]$ | `[2,9]` |
| 3 | $\left[\frac{3}{4},\ -y,\ \frac{1}{2}-y\right]$ | `[3,10]` |
| 4 | $\left[-y,\ y+\frac{1}{2},\ \frac{1}{4}\right]$ | `[4,21]` |
| 5 | $\left[y,\ y+\frac{1}{2},\ \frac{3}{4}\right]$ | `[5,23]` |
| 6 | $\left[\frac{1}{2}-y,\ \frac{3}{4},\ -y\right]$ | `[6,22]` |
| 7 | $\left[y+\frac{1}{2},\ \frac{1}{4},\ -y\right]$ | `[7,24]` |
| 8 | $\left[\frac{1}{4},\ -y,\ y+\frac{1}{2}\right]$ | `[8,13]` |
| 9 | $\left[y+\frac{1}{2},\ \frac{3}{4},\ y\right]$ | `[11,20]` |
| 10 | $\left[\frac{1}{2}-y,\ \frac{1}{4},\ y\right]$ | `[12,17]` |
| 11 | $\left[-y,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$ | `[15,19]` |
| 12 | $\left[y,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$ | `[16,18]` |
| 13 | $\left[\frac{3}{4},\ y+\frac{1}{2},\ -y\right]$ | `[25,38]` |
| 14 | $\left[\frac{1}{4},\ y+\frac{1}{2},\ y\right]$ | `[26,33]` |
| 15 | $\left[\frac{1}{4},\ \frac{1}{2}-y,\ -y\right]$ | `[27,34]` |
| 16 | $\left[\frac{1}{2}-y,\ y,\ \frac{3}{4}\right]$ | `[28,45]` |
| 17 | $\left[y+\frac{1}{2},\ y,\ \frac{1}{4}\right]$ | `[29,47]` |
| 18 | $\left[-y,\ \frac{1}{4},\ \frac{1}{2}-y\right]$ | `[30,46]` |
| 19 | $\left[y,\ \frac{3}{4},\ \frac{1}{2}-y\right]$ | `[31,48]` |
| 20 | $\left[\frac{3}{4},\ \frac{1}{2}-y,\ y\right]$ | `[32,37]` |
| 21 | $\left[y,\ \frac{1}{4},\ y+\frac{1}{2}\right]$ | `[35,44]` |
| 22 | $\left[-y,\ \frac{3}{4},\ y+\frac{1}{2}\right]$ | `[36,41]` |
| 23 | $\left[\frac{1}{2}-y,\ -y,\ \frac{1}{4}\right]$ | `[39,43]` |
| 24 | $\left[y+\frac{1}{2},\ -y,\ \frac{3}{4}\right]$ | `[40,42]` |

Table 10: Wyckoff site: `48j`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | `[1]` |
| 2 | $\left[x+\frac{1}{2},\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$ | `[2]` |
| 3 | $\left[x+\frac{1}{2},\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$ | `[3]` |
| 4 | $\left[z+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-x\right]$ | `[4]` |
| 5 | $\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$ | `[5]` |
| 6 | $\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$ | `[6]` |
| 7 | $\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$ | `[7]` |
| 8 | $[x,\ -y,\ -z]$ | `[8]` |
| 9 | $[-x,\ y,\ -z]$ | `[9]` |
| 10 | $[-x,\ -y,\ z]$ | `[10]` |
| 11 | $\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$ | `[11]` |
| 12 | $\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$ | `[12]` |
| 13 | $\left[\frac{1}{2}-x,\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$ | `[13]` |
| 14 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$ | `[14]` |
| 15 | $\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ x+\frac{1}{2}\right]$ | `[15]` |
| 16 | $\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$ | `[16]` |
| 17 | $[z,\ x,\ y]$ | `[17]` |

*continued ...*

Table 10

| No. | position | mapping |
|---|---|---|
| 18 | $[y,\, z,\, x]$ | [18] |
| 19 | $[-y,\, z,\, -x]$ | [19] |
| 20 | $[-z,\, -x,\, y]$ | [20] |
| 21 | $[-y,\, -z,\, x]$ | [21] |
| 22 | $[z,\, -x,\, -y]$ | [22] |
| 23 | $[y,\, -z,\, -x]$ | [23] |
| 24 | $[-z,\, x,\, -y]$ | [24] |
| 25 | $\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$ | [25] |
| 26 | $[x,\, -z,\, y]$ | [26] |
| 27 | $[x,\, z,\, -y]$ | [27] |
| 28 | $[z,\, y,\, -x]$ | [28] |
| 29 | $[-z,\, y,\, x]$ | [29] |
| 30 | $[-y,\, x,\, z]$ | [30] |
| 31 | $[y,\, -x,\, z]$ | [31] |
| 32 | $\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$ | [32] |
| 33 | $\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$ | [33] |
| 34 | $\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$ | [34] |
| 35 | $[y,\, x,\, -z]$ | [35] |
| 36 | $[-y,\, -x,\, -z]$ | [36] |
| 37 | $[-x,\, z,\, y]$ | [37] |
| 38 | $[-x,\, -z,\, -y]$ | [38] |
| 39 | $[z,\, -y,\, x]$ | [39] |
| 40 | $[-z,\, -y,\, -x]$ | [40] |
| 41 | $\left[z+\frac{1}{2},\, x+\frac{1}{2},\, y+\frac{1}{2}\right]$ | [41] |
| 42 | $\left[y+\frac{1}{2},\, z+\frac{1}{2},\, x+\frac{1}{2}\right]$ | [42] |
| 43 | $\left[\frac{1}{2}-y,\, z+\frac{1}{2},\, \frac{1}{2}-x\right]$ | [43] |
| 44 | $\left[\frac{1}{2}-z,\, \frac{1}{2}-x,\, y+\frac{1}{2}\right]$ | [44] |
| 45 | $\left[\frac{1}{2}-y,\, \frac{1}{2}-z,\, x+\frac{1}{2}\right]$ | [45] |
| 46 | $\left[z+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-y\right]$ | [46] |
| 47 | $\left[y+\frac{1}{2},\, \frac{1}{2}-z,\, \frac{1}{2}-x\right]$ | [47] |
| 48 | $\left[\frac{1}{2}-z,\, x+\frac{1}{2},\, \frac{1}{2}-y\right]$ | [48] |